

Sep 15, 2026

Virtual Gold x CMU

Capstone Sync

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Attachments  Virtual Gold x CMU Capstone Sync

Summary

Project scope review with hybrid architecture evaluation and domain selection discussion

Hybrid Architecture and Research

Discussion covered local confidence scoring and proposed hybrid routing architecture. Evaluation metrics included accuracy and latency.

Domain Selection and Use Case

Debate focused on financial services versus retail domains. Consideration was given to horizontal application approaches.

Administrative Details and Setup

Review of timelines and cloud storage management. Agreement was reached on setting up shared document folders.

Decisions

Needs Further Discussion

- **Project use case domain selection** The group deferred finalizing whether the project will pursue a vertical financial services use case or a horizontal sensitive-data sanitization application until the next meeting.

Aligned

- **Shared live document storage** The team aligned on creating a shared folder in OneDrive or Google Drive to host a live document for collaborative editing instead of circulating multiple file versions.

Next steps

- ☐ [The group] Finalize Project Use Case: Select between the financial trading application or a horizontal data sanitization solution. Discuss and confirm the direction to guide the scope of work.
- ☐ [The group] Research Redaction Tools: Investigate existing open-source or commercial software for sensitive information removal. Identify suitable components to support the horizontal application strategy.
- ☐ [The group] Research Financial Data: Identify available market data and corporate filings relevant to the trading app proposal. Determine accessibility and sources for these datasets.
- ☐ [Inderpal Bhandari] Propose Use Cases: Provide feedback and specific scenario recommendations for the project. Send suggestions for doable tasks within the 15 week timeline.
- ☐ [The group] Address Slack Budget: Contact Ruth regarding the upcoming expiration of the free Slack plan. Determine the procedure for upgrading to the paid tier to avoid service disruption.
- ☐ [Mark Chu] Check Subscription: Check if she already has a subscription for Slack.
- ☐ [The group] Discuss Issue: Evaluate the main issue and research potential solutions.

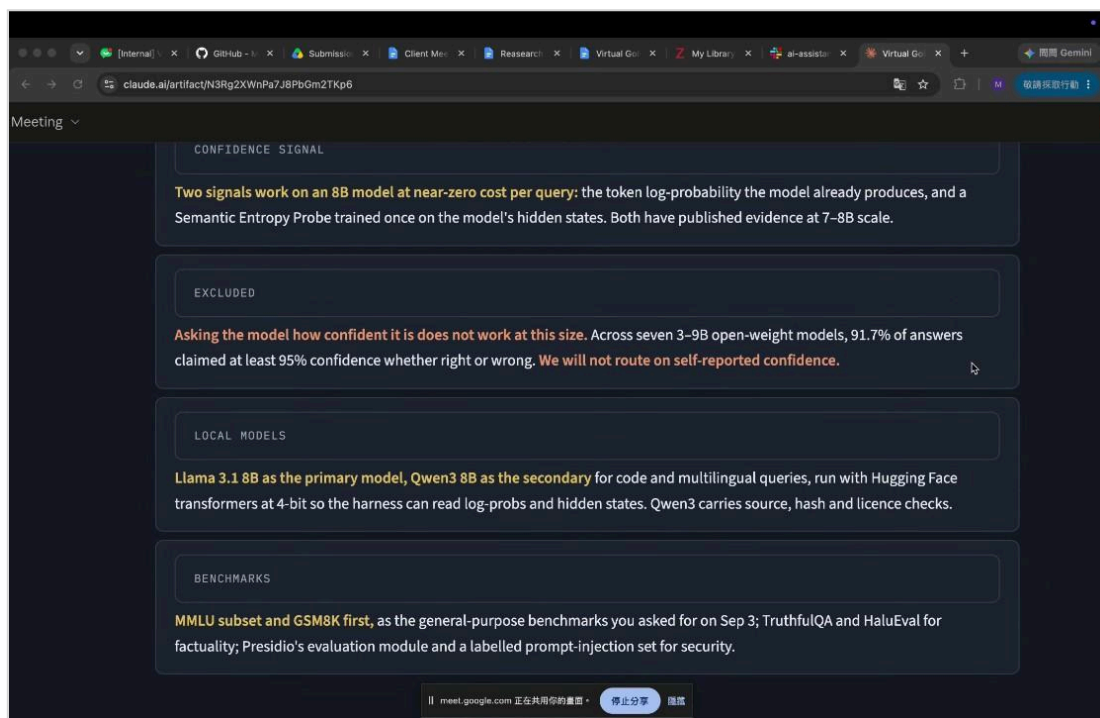
- ☐ [The group] Determine Scenario: Specify if the target application scenario is horizontal or vertical.
- ☐ [Inderpal Bhandari] Send Thoughts: Send personal perspectives on the topic to the team.
- ☐ [Inderpal Bhandari] Setup Folder: Create a OneDrive location for shared live documents.
- ☐ [Mark Chu] Share Architecture: Provide diagrams showing sensitive information handling within the architecture.
- ☐ [Alex] Review Architecture: Evaluate specific blocks related to sensitive information parsing in the architectural diagram.

Details

- **Project Scope and Agenda:** Mark Chu opened the week four meeting by noting that the team has 15 weeks remaining to complete the hybrid AI assistant capstone project. Mark Chu outlined a five-part agenda covering architectural research, architecture review, benchmark comparisons, open questions regarding credits and sandboxes, and the scope of work agreement.

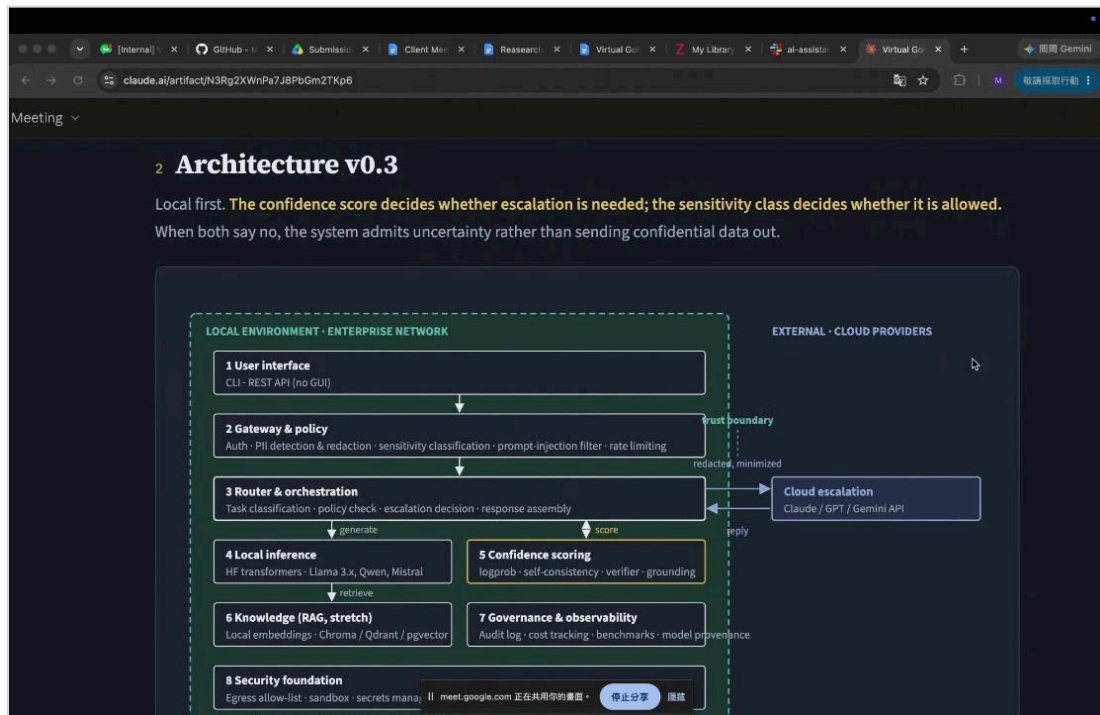
The screenshot shows a web application interface with a dark theme. At the top, there's a navigation bar with several tabs: [Internal], GitHub, Submission, Client Meeting, Research, Virtual Gold, My Library, AI Assistant, Virtual Gold, and Gemini. The main header area includes the text 'CMU CAPSTONE · VIRTUAL GOLD INC · CLIENT MEETING 2 OF THE DISCOVERY PHASE' and a large title 'Research, architecture, comparison plan, and the Scope of Work'. Below the title, there are two sections: 'DATE' (Monday, September 15, 2026 · Week 4 of 15) and 'GOAL' (Agree on scope and resources, then sign the Scope of Work). A legend indicates: Gold: key finding or decision, Blue: needs your answer today, Red: excluded or deferred. The main content area features five numbered cards: 1. Research (what we read, what we chose), 2. Architecture (v0.3, local-first), 3. Comparison (three configurations, six metrics), 4. Open questions (sandbox, credits, sector, defaults), and 5. SOW signature (fill the cost line, sign). Below the cards, there's a section titled '1 Research based on the project description' with a paragraph of text: 'The brief asks for a hybrid assistant that escalates to the cloud "when additional confidence or reasoning capability is required". We researched what that cor...'. At the bottom, there's a footer with a Google Meet link and a '停止分享' button.

- Research on Confidence Scoring and Model Selection:** Mark Chu presented research on competence signals and harness policies, highlighting token log probability and semantic entropy as near-zero-cost query methods. Mark Chu explained that the team avoids asking models directly for their confidence because open-weight models between 3 and 9 billion parameters claim at least 95% confidence 91.7% of the time, resulting in saturation. For local models, they plan to use Llama 3.1 8 billion as the primary model and a 3 billion parameter model for code and multilingual queries, starting with MMLU subsets and GSM8K benchmarks.

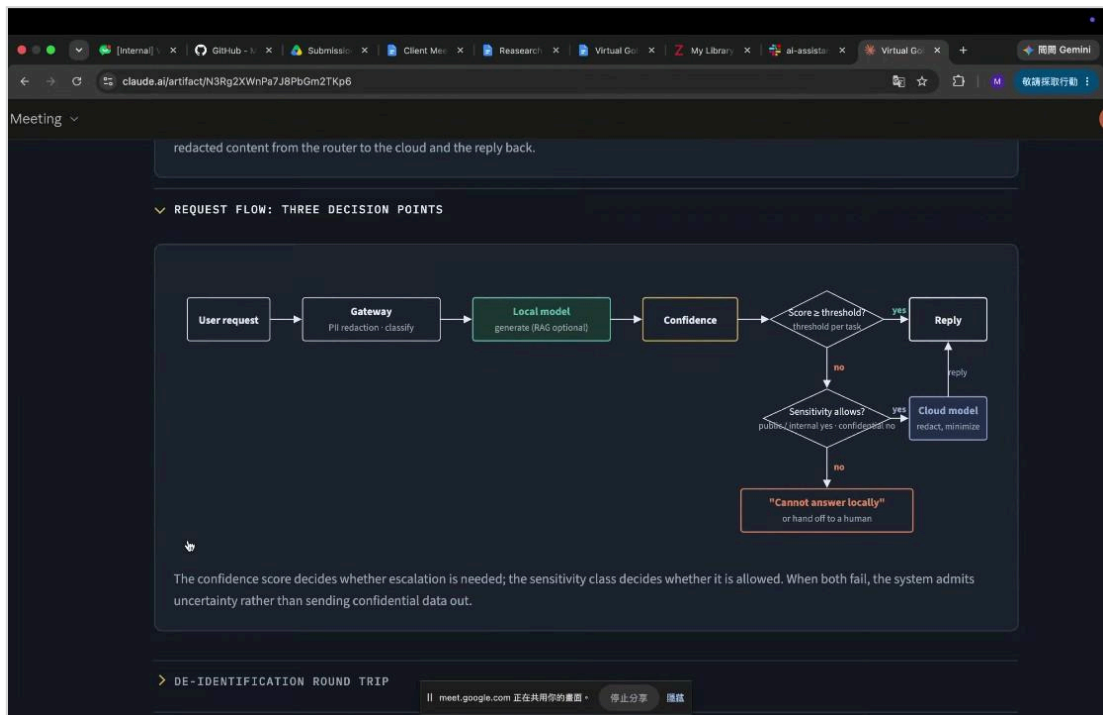


- Evaluation of Confidence Methods and Benchmarks:** Alex asked for clarification on how confidence levels and benchmarks interrelate. Mark Chu explained that benchmarks are utilized to evaluate different methods before adopting a final competence signal, and that the local model's confidence score serves as evidence to determine whether to route API calls locally or to cloud APIs.
- Proposed Hybrid Architecture Overview:** Mark Chu presented the proposed local-first hybrid architecture featuring a REST API without a graphical user interface, a gateway handling policy authentication and personal identifiable information detection and reduction, a custom router and orchestration layer, and local inference using Hugging Face transformers. Mark Chu detailed that the architecture uses confidence

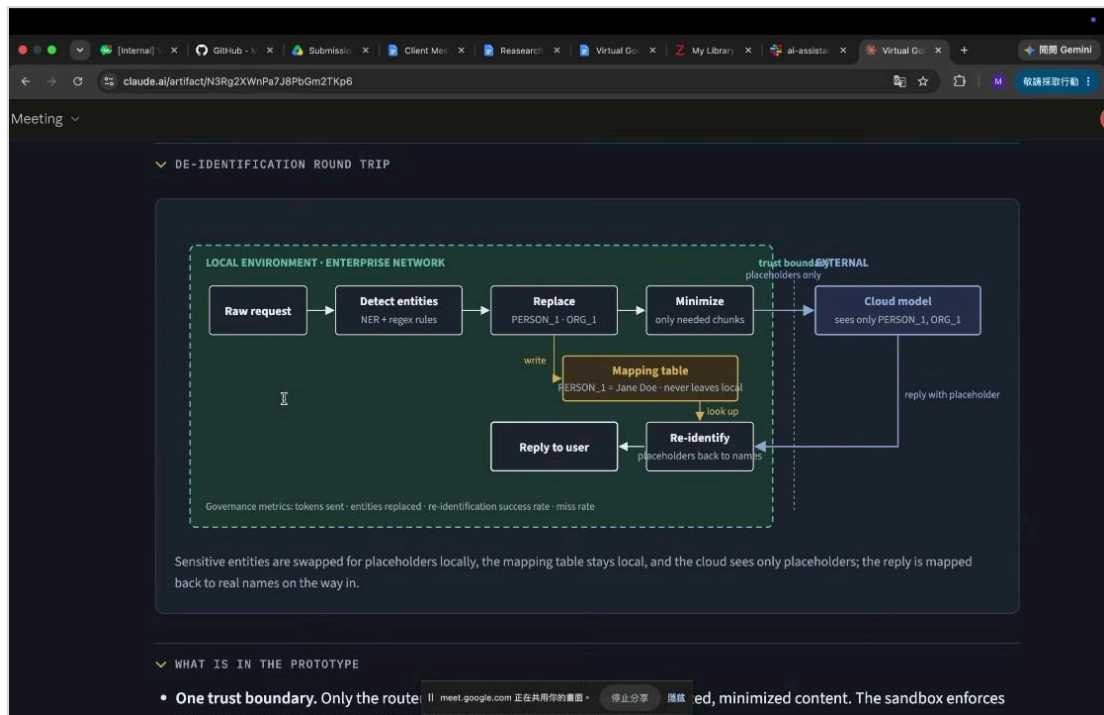
scores to decide if cloud escalation is necessary and least-privilege classes to regulate data transmission to the cloud.



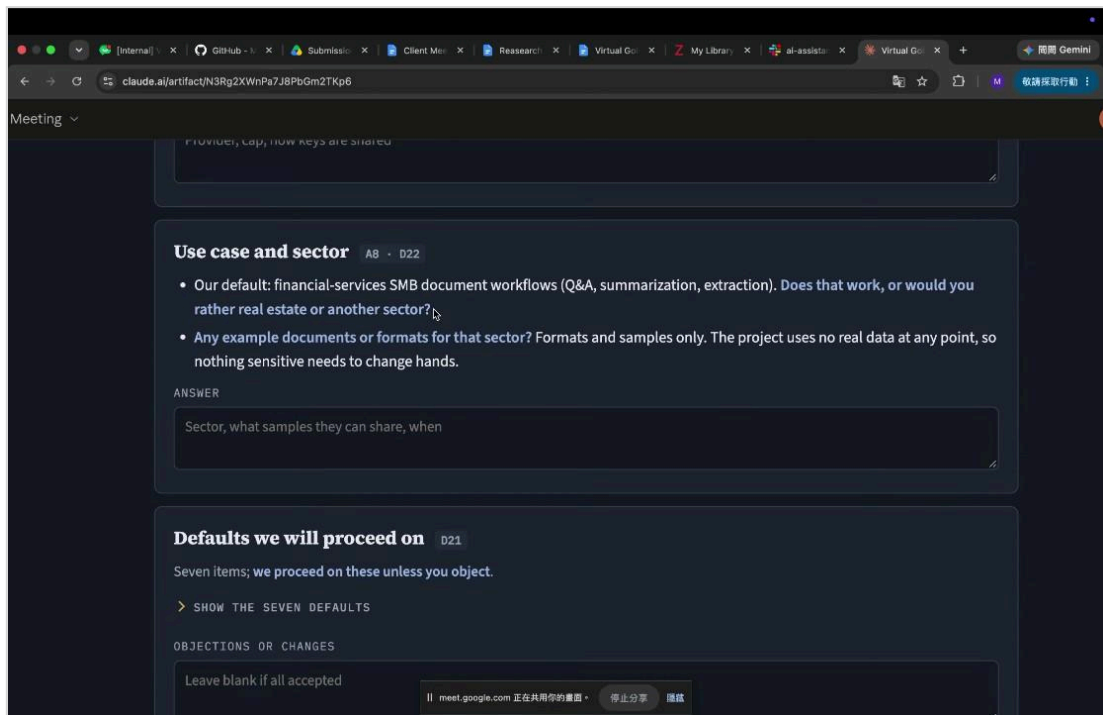
- **Sensitivity and Data De-identification in Architecture:** Alex questioned how information sensitivity is determined and suggested considering partial prompt restriction, output censorship, or split routing instead of a strict binary gate. Mark Chu noted that a separate de-identification component aims to remove sensitive information before sending data to cloud APIs, while Anmol Sahu added that the team is exploring data tagging as internal or confidential.



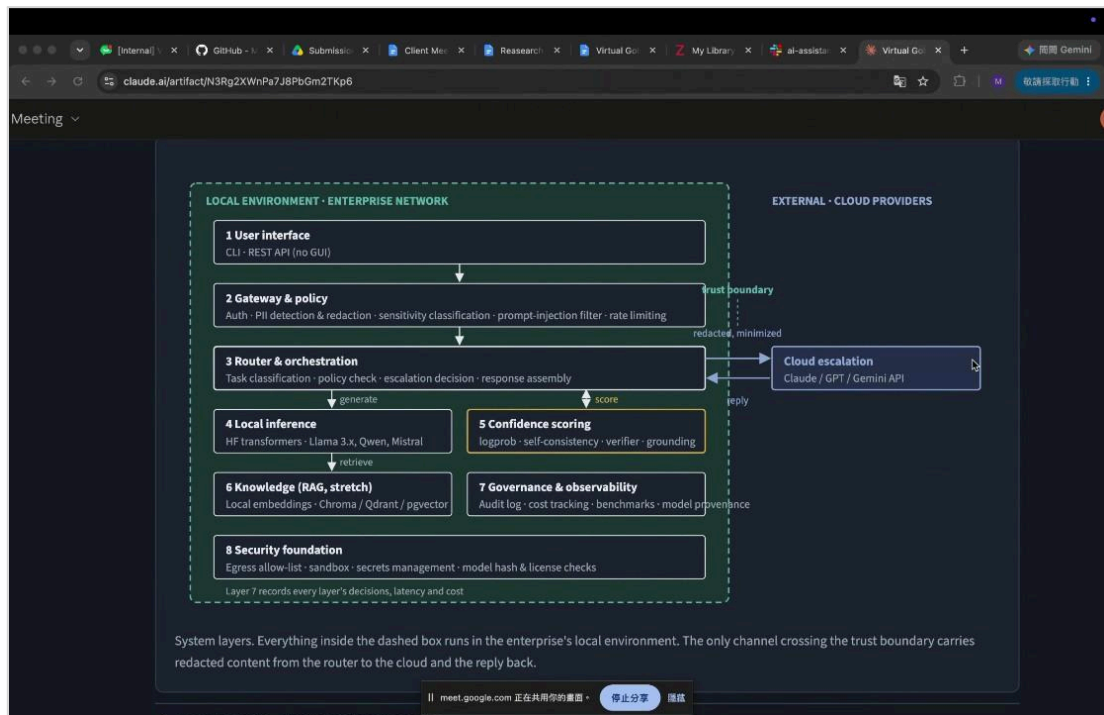
- **Upstream Systems and Data Pipeline Handling:** Priya Iragavarapu inquired about the upstream and downstream systems feeding the local layer. Anmol Sahu responded that the high-level architecture is designed to be scenario-agnostic, while Mark Chu explained that confidential data can be processed via a data pipeline that replaces customer and organization names or IDs with random numbers before cloud transmission, followed by re-identification in a local loop.



- **Selecting a Domain and Use Case for the Project:** Priya Iragavarapu suggested anchoring the problem statement around a specific domain such as retail, while Mark Chu proposed focusing on financial services like credit cards, loans, and banking accounts to facilitate synthetic testing. Inderpal Bhandari subsequently suggested a specific financial trading use case utilizing publicly available market data and company filings, allowing traders to keep trading strategies local while leveraging market intelligence.



- **Implementation Platform and Timeline Feasibility:** Inderpal Bhandari questioned the implementation platform and whether the project is doable within 15 weeks. Mark Chu confirmed the focus on open-source local models like Llama and rebuilding LightLLM by hand for the gateway to ensure data privacy and transparency, while Anmol Sahu stated that the timeline is fully achievable and accelerated by the use of AI coding agents like Claude or Codex.



- **Configuration Scenarios and Evaluation Metrics:** Mark Chu proposed testing three configurations—local only, hybrid, and cloud only—evaluated across metrics including accuracy, latency, cost per query, escalation rate, personal identifiable information leak rate, and prompt injection resistance. When Priya Iragavarapu asked how accuracy is assessed, Anmol Sahu proposed creating a golden dataset or using an LLM as a judge, prompting Priya to note that pre-published mock datasets with ground truth are freely available online.

Meeting

3 Comparison: how we will show the trade-off

The same test set runs under three configurations. That is the only way to answer your two questions from Sep 3: **how big is the gap to the cloud, and what does the local path save.**

CONFIGURATION	WHAT IT IS	WHAT IT TELLS US
Local-only	Every request answered by the 8B model, no escalation	Quality floor, cost floor, privacy ceiling
Hybrid	Local first; confidence and sensitivity decide escalation	Where our proposal lands between the two extremes
Cloud-only	Every request straight to the cloud model	Quality ceiling, cost ceiling, privacy floor

Six metrics per configuration: accuracy · latency · cost per query · escalation rate · PII-leak rate · prompt-injection resistance.

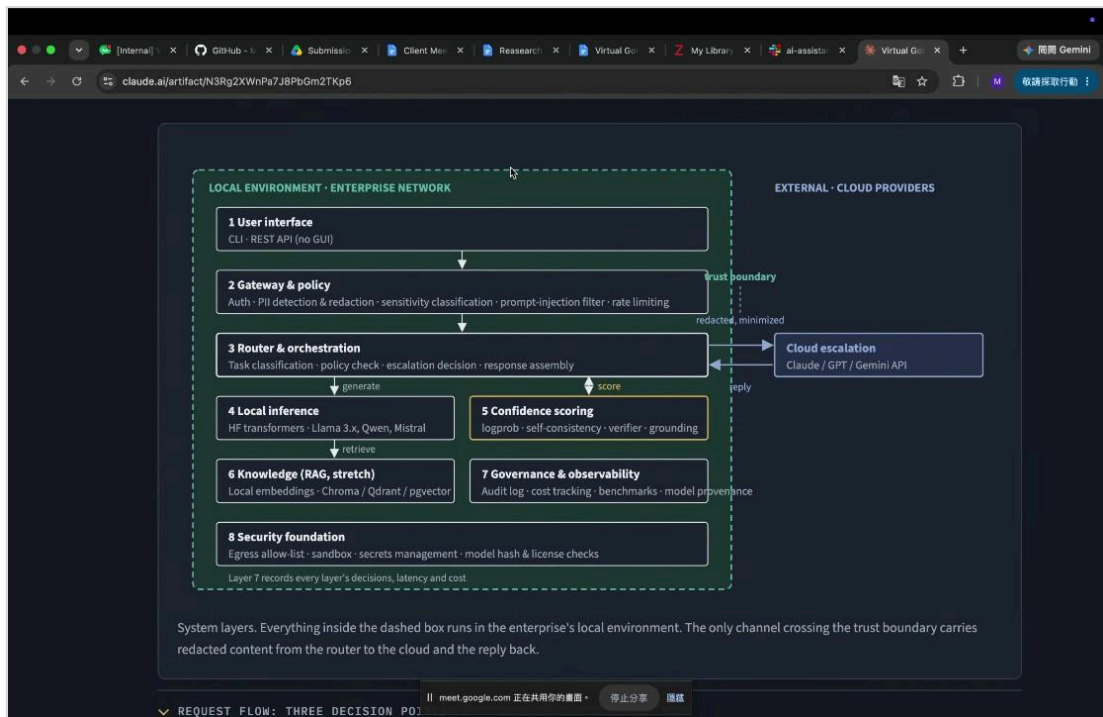
WHAT YOU WILL SEE AT THE MIDPOINT (OCT 2-9, WEEK 7)

THREE CONFIGURATIONS, SIDE BY SIDE

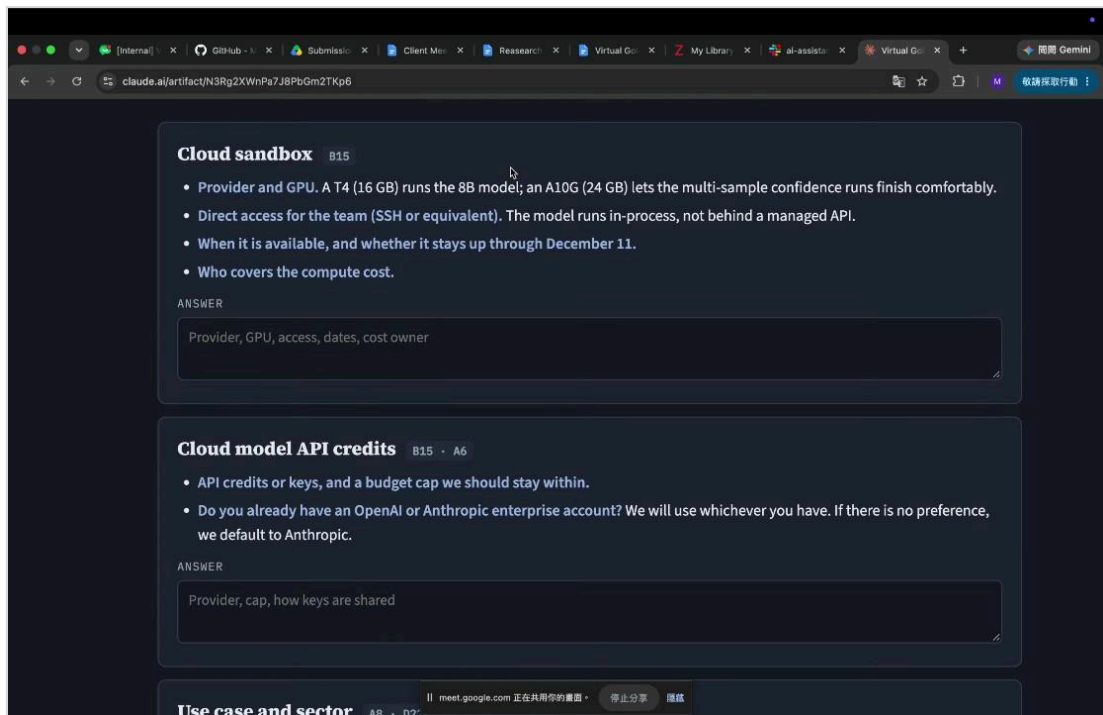
Local-only, hybrid and cloud-only, each run on the same MMLU subset and GSM8K questions, so the gap between the 8B model and the cloud model is measured

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- Horizontal versus Vertical Application Approaches:** Alex and Inderpal Bhandari discussed whether the project should target a specific vertical use case or a horizontal application such as sanitizing confidential data across multiple industries. Anmol Sahu clarified that the gateway policy and personal identifiable information detection module could be developed as a horizontal application, while the complete architecture functions as a vertical application, with Inderpal noting that implementing a horizontal capability still requires an underlying architecture.



- **Next Steps, Resources, and Administrative Details:** Mark Chu inquired about API credits and sandbox environments, to which Inderpal Bhandari confirmed that OpenAI and Anthropic accounts and open-source frameworks are available for the team to leverage. Mark Chu noted that the scope of work agreement remains pending until the application scenario is finalized, and concluded by mentioning that the team's Slack free plan expires in 30 days and will require a \$15 monthly upgrade, which Inderpal asked if Ruth had assisted with.



- Slack Subscription and Group Discussion Goals:** Inderpal Bhandari mentions that someone may already have a Slack subscription and asks Mark Chu to verify this status. Mark Chu states that their group will discuss the issues previously raised by Inderpal Bhandari, Alex, and Priya Iragavarapu, work toward a solution, provide new research, and determine whether the application scenario is horizontal or vertical.
- Timeline for Finalizing Topics and Sharing Thoughts:** Inderpal Bhandari suggests trying to finalize the topic within the week so that everyone is prepared for the upcoming meeting, and commits to sending out their thoughts. Mark Chu agrees to work quickly to share their group's topic and thoughts in advance to enable a more efficient discussion.
- Collaborative Document Management via Cloud Storage:** Priya Iragavarapu recommends creating a folder in OneDrive or Google Drive to host a live shared document, avoiding the circulation of multiple versions and allowing participants to add comments and synthesize a final version collaboratively. Inderpal Bhandari and Mark Chu agree with the approach, and Inderpal offers assistance with setting up OneDrive.
- Architecture Diagram and Sensitive Information Flow Review:** Alex requests a review of the previously shown diagram and architecture once the topic and focus are established. Specifically, Alex wants to examine how the system determines what constitutes sensitive information and how that

information is parsed out in the workflow. Mark Chu acknowledges and confirms the plan.

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